Simulation and Optimization of UTSA Campus Chick-fil-A

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Abstract

Chick-fil-A is a staple for students across UTSA campus and serves hundreds of students each day. Whether they are rushing to their next class or taking a quick study break, students expect a fast and efficient experience. As one of the most popular restaurants with a limited amount of space, wait time and queue length are growing to a point where it is no longer a convenient option. Using advanced manufacturing concepts and simulation software, the current setup was simulated and analyzed. Following the initial analysis, an optimized system was presented and simulated for comparison purposes. Results showed that an increase in resources decreased total system time average by 61.92%. Further discussion and recommendations are suggested.

1. Introduction

Stopping operations to strategically find solutions to continue meeting customer demand is not always a viable option; simulation modeling is a key tool for analyzing performance, as it provides a safe way to test and investigate different scenarios while avoiding interference with your current operations for service customers. Simulation helps solve real-world problems safely and efficiently in a way that is easily confirmed and communicated to provide clear alignment for all stakeholders. When running a restaurant in the service industry, simulation provides for quick and economical measure for gauging wait times for customers, throughput, and queue length experienced by the system. Simulation allows for complicated interactions to be solved more easily than by only utilizing mathematical models. This not only saves time and money for those involved, but also provides increased accuracy through being able to simulate with many more variations and details that go into a real-life scenario.

1.1 System Description

Founded in 1946, Chick-fil-A, Inc. owns and operates franchises offering quick-service chicken restaurants in the United States. The many different types of franchises include, but are not limited to, stand-alone restaurants, drive-thru only outlets, full-service restaurants, non-traditional outlets, satellite branches and also in-mall or in-line restaurants. As part of their in-line restaurants, the University of Texas at San Antonio added a franchise located on the first floor of the John Peace Library with an abbreviated-menu. This abbreviated menu would include the most popular items like the chicken sandwiches and chicken nuggets as well as their famous waffle fries.

It's these special food items that drive customers to Chick-fil-A on a daily basis. When you incorporate a young population, mainly made up of young adults aging from 18-24, this business is expected to thrive on a daily basis with an ever-constant demand. The demand experienced by this Chick-fil-A restaurant is steady, yet somewhat unpredictable. Busy or peak times take place at lunch and dinner time as expected but also during the times between classes based on the class schedule set by the university. This creates variability and randomness in the arrival rates for both busy and slow times experienced by this restaurant as customers are normally in a hurry to pick up food and on to their next activity.

The exact restaurant under consideration is located in the food court at UTSA and is built in such a way that food is made available for customers to simply arrive, grab their food, and pay. The restaurant uses stanchions to create an organized line for the customers to follow when making their way through the system. When the customer arrives to the restaurant, they pass through the stanchions to the first service location where they can grab their desired food

item. If a customer wants a custom food item that is not available at the food counter, he/she waits in line to order from the employees preparing the food. This customer usually waits until his item is ready and then reenters the checkout line. Customers do not wait behind this person but skip ahead in the line and proceed to the checkout. The checkout counter is typically ran by one service employee but if the system is busier than normal, two service employees are available to run the checkout counters. The service employees not only process payments but also hand out condiments, drinks, and paper bags when asked for. After the customer has been served, he or she leaves the system and the next customer moves into the checkout process.

1.2 Observation of the System

After observing the current setup of Chick Fil-A**,** the bottlenecking of customers occurred in two separate locations. The first location was immediately after the customer entered the restaurant approaching the food service counter and the second location occurred after the customer grabbed their food and made their way to the checkout station. Each of these bottlenecks increased the customer wait time. Also, it was noticed that when customers entered the food court, some customers would decide not to enter the restaurant due to the long line of customers. Therefore, reducing customer wait time would lead to an increase in business for the restaurant. Two ways to decrease the customer wait time and reduce the bottleneck is to configure each of the service stations within the model. The first suggestion would be to add a second food service counter. This implementation would increase the amount of food readily available for customers as they would have a decision of choosing food from two separate locations. The second suggestion would be to accompany the additional food service counters with an extra checkout service station. The additional checkout would assist the restaurant in processing the customers payments for the food and decreasing the bottleneck before the customer departs from the restaurant.

1.2.1 Assumptions

To test and analyze the effects these changes would have on the system described above, two simulation scenarios were designed and created. The most important metric of the simulation is the average time the customer spends inside of the system, that is from the time that they arrive to the time that they pay and leave. To accurately model the system, some simplifying assumptions have been made:

1. The cashiers are treated as identical, with the same tasks and abilities and the same service rates.
2. Data collection was limited to Tuesday and Thursday, 4:00 pm to 6:00 pm, and is used to represent all peak hours on weekdays.
3. Simulation was focused on only peak hours and therefore scenarios considered may not apply to less busy times.
4. Chick-fil-A, for this simulation purpose, is ‘open’ for six hours a day, which represents the collective duration of peak hours.
5. There are no shifts between the workers.
6. There are no breaks for the workers during the time when the model is running.
7. Customers who enter the line and exit before purchase are not considered
8. When the customer ‘leaves the store’ this implies their transaction is complete and their food has been received.
9. The food counter, special order counter and check out are considered service stations.

2. Collected Data

The data collected to help properly simulate Chick-fil-A during peak and slow hours include customer arrival rates, service time in the food service area, as well as cashier checkout rate. Time data was collected and recorded in seconds through an excel tool utilized by a professional services firm for time study observations.
The first data set was collected for two hours on a Thursday evening, when customer arrival rate was considered slower than usual. The team recorded the following time stamps: Customer arrival, customer grabs ready item or orders custom item, customer enters checkout queue, customer reaches cashier, and customer leaves restaurant. Each time an event occurred; it was noted in the time study tool by the push of a button.

The second set of data was recorded on a Tuesday evening for two hours during the dinner rush. This data was to be used to find rates used for ‘busy’ hours. The same procedure as previously described was used to collect these times.

This data provided all time stamps necessary to deduct customer arrival rate, food counter service rate, and cashier service rate. Following observation, the data was then prepared for the input analyzer, which will be discussed in the following section.

2.2 Data Processing for Distribution Analysis

Following the time study, it was necessary to take the raw data of inter-arrival times and service times and fit it to a distribution to be used as an expression within Arena. Creating these expressions provided more accurate service and arrival times for our resources. To prepare the raw data, inter-arrival times were first saved into separate text files. These text files were then imported into the Input Analyzer software and ran find the correct distribution fit. The Input Analyzer works by taking the raw data, creating a histogram graph, and then fitting to the graph to provide a distribution summary used for the current system modules. This process is illustrated with Figures 1-3.

Arrival Rate Expressions

Slow Session Busy Session

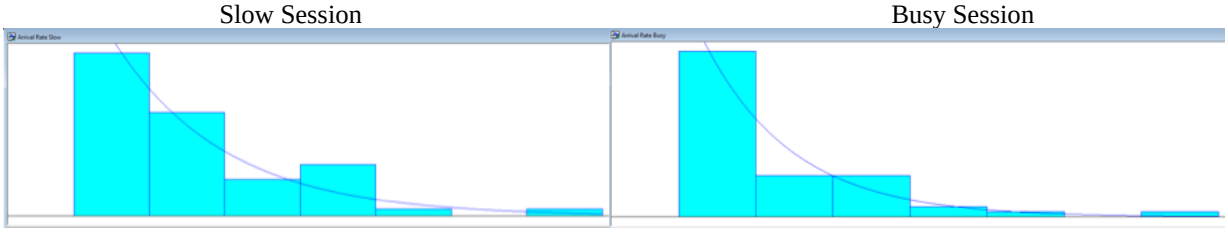

Figure 1: Arrival Rate Distribution Fits

Order Processing Rate Expressions

Special Order Ready Order

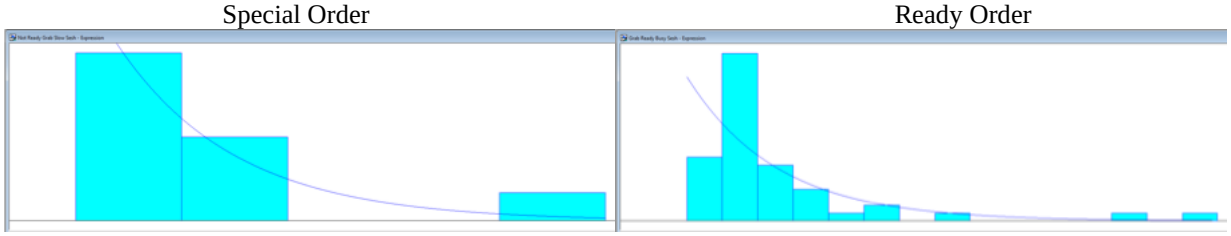

Figure 2: Order Processing Rate Distribution Fits

Cashier Processing Rate Expressions

Slow Session Busy Session

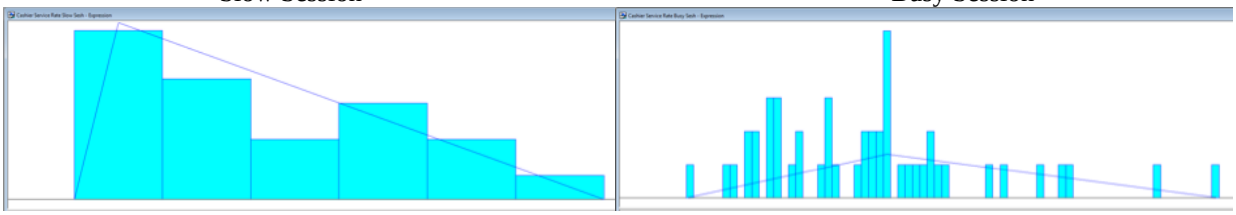

Figure 3: Cashier Processing Rate Distribution Fits

3. Current System Simulation

3.1 Simulation of Current Setup

To effectively model the current setup of Chick-fil-A, it was necessary to first break down the customer's path from start to finish. The procedure each customer follows is shown below in Figure 4 and will be further translated into various modules within Arena.

Figure 4: Chick-Fil-A Process Flow Chart

From this logic, modules such as Create, Process, Record, Assign, Decision, etc. will be used to simulate the current setup. Table 1 below is a summary of the model's parameters and their corresponding variable of Chick Fil A.

Table 1: Summary of Chick-Fil-A Parameters

	Arena	Chick-fil-A	Type/Action
1.	Entities	Customers	Part
2.	Resource 1	Food Counter Clerk	Seize Delay Release
3.	Resource 2	Custom Order Clerk	Seize Delay Release
4.	Resource 3	Cashier	Seize Delay Release
5.	Queue 1	Food Queue	FIFO
6.	Queue 2	Custom order Line	FIFO
7.	Queue 3	Checkout Line	FIFO
8.	Decision 1	Custom Order or Grab Ready-Made Item	2-Way by Chance

Figure 5 displays the complete Arena model of the current setup.

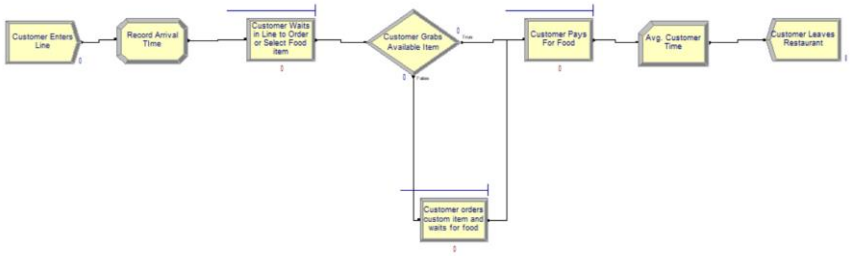

Figure 5: Arena Model of Current System

The construction of the model will be explained step by step to aid in the understanding of the logic behind simulation.

First, to simulate the customer entering the restaurant, a Create module is placed and named ‘Customer Enters Line’. This module also requires an expression for arrival rate, which was found previously using the input analyzer. The data was fit using an exponential distribution as it helps to describe the time between different number of events inside of a Poisson Process, such as the arrival of customers into the restaurant. The modules dialog box is shown below and shows the expression use along with the units of measurement.

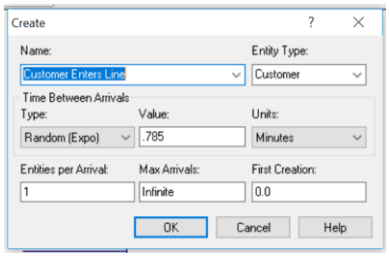

Figure 6: Create Module Dialog Box

Next, the customer waits in line to reach the food counter where they will eventually make their decision to either grab a ready item, or order a custom item with the food clerk. The food counter is displayed as a process module with a queue holding the customers waiting to reach that point. This process module has the action ‘Seize Delay Release’ because the food counter is seized by customer, delayed by their action of grabbing food, and then released. The expression used is from the data observed and is the difference between when the customer enters the line and reaches the food counter to grab his/her food. Because the food counter acts as a self-service center for the customer it was decided to use a triangular distribution to accurately simulate the process. The triangular distribution uses the maximum and minimum time a customer takes to grab the food along with the average, to create the simulation of customers waiting in queue at this moment. The process module is shown below in Figure 7:

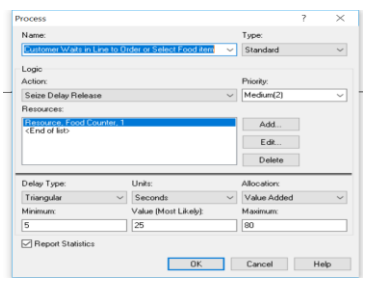

Figure 7: Process Module Dialog Box

From the food counter step in the process, it was observed that if a customer decided he/she did not want an already prepared food item, they must still pass through the food counter station and then request for the alternative choice in the middle of the line. To emulate this process in the model, the probability of when a customer chooses to make a “special” order was calculated based on the number of times a customer made this specific decision. Using a decision module, with a true or false function, customers who did not want a special order simply moved on to the checkout counter, while the remaining who did make a special order stayed in line until their order was prepared. The decision module and process module for when the customer makes their special order are shown in Figure 8 below.

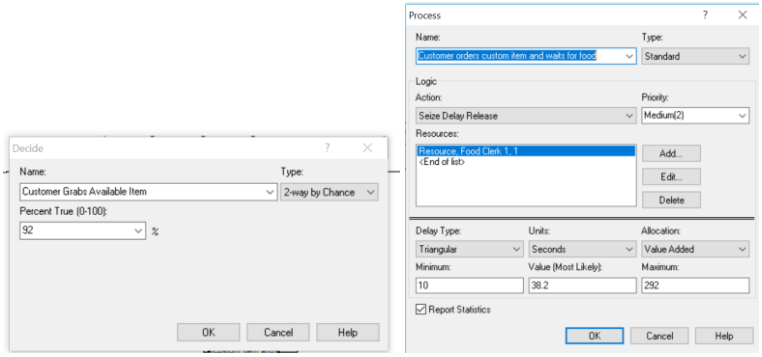

Figure 8: Special Order Decision and Process Module Dialog Boxes

As shown in the decision module, the probability of a customer not making a special order was 92%, leaving a remainder of 8 % of customers stopping inside of the line to make their special order. A triangular distribution was used along with a seize delay release type logic as the customer is being served by the food preparation server of the restaurant. It was assumed that if a customer decides to make a special order they are skipped in line and reenter once their request is prepared and exchanged.

The last step in the process is the checkout counter where the customer pays for food. A process module was used here and is shown below with each of the values entered for the triangular distribution.

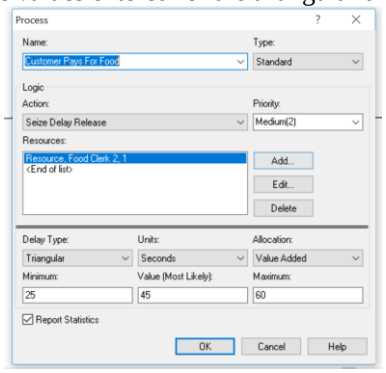

Figure 9: Checkout Counter Process Module Dialog Box

The times used for each of the distribution values were calculated by finding the difference between when customers enter the checkout counter and when they leave. This process concludes the model and the customers move to a departure model meaning that they leave the system. The results of the model were exported into a report and include several measures of performance that are discussed in the following section.

3.2 Output Analysis of Current Setup

The model was setup to run 150 replications with a one hour warm up period for a duration of 6 hours. The large number of replications will ensure that our confidence interval is not too wide as well as limit our variance. Peak hours were estimated as 6 hours a day during operational hours, for this purpose a one-hour warm up period was included to account for the restaurant being operational prior to targeted times. Following all replications of the current system model, Arena provides a summarized PDF of parameter averages, minimums, maximums, etc. This summarized report will be used to analyze the performance of the current setup as well as compare to the improved setup.

The average total number of customers recorded through the system was 451, this value breaks down to about 75 people an hour. Based on our time study, the realistic number of arrivals during peak hours was 60 people in one hour. This verifies that the arrival rate is a good representation of reality.

The first important metric to be evaluated is overall time spent within the system. The amount of time each student spends at Chick-fil-A is the objective to be minimized and was found to have a maximum value of 35 minutes, and an average of 5.22 minutes. A closer look will be taken into what queue time is contributing to this maximum value of 35 minutes.

As described previously, there are three queues within the system: Customers waiting to order or grab a ready item, customers waiting for a custom food item, and customers waiting to checkout. Average and maximum values for number of customers waiting in queues and waiting time are summarized below in Table 2.

Table 2: Summary of Queue Wait Time and Length Data

Queue	Waiting Time (Minutes)		Queue Length (Customers)	
	Average	Maximum	Average	Maximum
Waiting to Order	1.248	19.354	1.609	32
Waiting on Custom Item	0.249	8.737	0.026	3
Waiting to Checkout	2.484	25.423	3.215	37

The final parameter used to determine performance was the utilization of resources. At each process module containing a queue, there was a resource: Two food clerks manning the food counter and taking custom orders, and the cashier taking payments. The average utilization was highest for the cashier at 0.9058 and followed by the food counter clerk and custom order clerk at 0.7770 and 0.1931, respectively. This data makes sense, as only 8% of customers are ordering a custom item, and all customers must wait to pay and there is only one cashier available.

With this data, it can be concluded that this is not the ideal setup for peak hours; an average student would not have time to wait in line for the maximum 35 minutes. Even an average time of 5.22 minutes could be lowered significantly, as this restaurant is setup to be a 'grab and go' option which takes a minimal amount of time. The goals when redesigning the model, will be to decrease overall wait times and queue length in each queue, as well as distribute the utilization of resources in a way that does not create such a bottleneck at the checkout counter.

4. Chick-Fil-A 2.0 Simulation

4.1 Changes Implemented

After examining the results of the current model, it was noticed that the overall wait times and queue lengths of each queue were large in length. In order to combat these high wait times, it was decided to focus on the optimal location for each of the service stations within the system. In this case, the food counters, special order area, and checkout counters are all considered service stations. A second model of Chick-fil-a was constructed in Arena which is shown below in Figure 10.

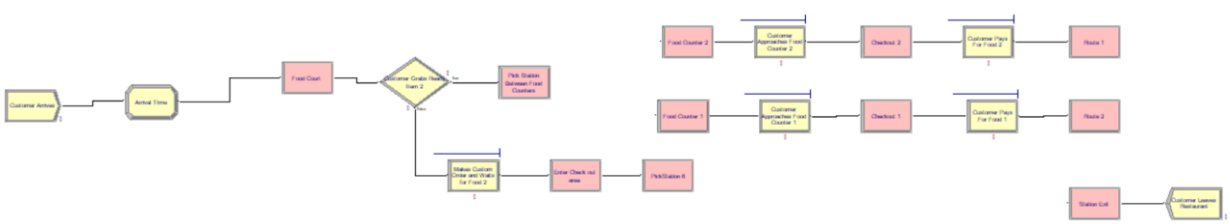

Figure 10: Second Model of Chick-Fil-A

The system above is a hypothetical model assumed to be the best way of decreasing customer wait times, total time in the system, and queue lengths. The data and expressions used through the new system remain the same for each of the processes as they were in the real-life model. Using advanced process modules in arena the model was able to be structured to incorporate multiple food and checkout counters. At the beginning of the process the customers enter the system like before where the interarrival times are measured by an exponential distribution.

The customers move to a station module which simulates the customers entering the restaurant. The decision module used for whether or not the customer wants to place a special order, was moved to the beginning of the process before entering a line. When the customer makes his decision, he/she has the choice of entering two separate lines.

Each of the two lines includes a food counter and a checkout counter at the end. A pick station module was used to model customers choosing which line they want to go to. The module was set up in such a way that customers would select the line based on the queue length of the two food counters, meaning that if one food counter was too crowded the customer would select the other line. The pick station module dialog box is shown below to convey the parameters selected.

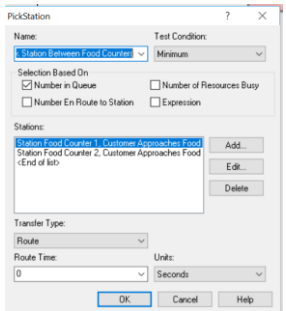

Figure 11: PickStation Module Dialog Box

If a customer did decide to make a special order, he/she would move to a separate location where employees would be there to take their order. After their order was a complete a second pick station module was used to allow the special-order customers to select then which checkout line they would like to enter. These check outlines involve the customers that selected food from the food counter.

Once the customers have selected which food counter they would like to use, the customers are then fixed to the same route for the checkout line. Because the real life model exposed the food counters and checkout lines to be the greatest potential bottleneck it is hypothesized that the restructure of the model will eliminate the bottle necks and simultaneously decrease customer wait time at the service processes, total time in the system, and the queue lengths for each of the service processes.

4.2 Output Analysis of Revised System

After implementing changes, the second model was run using the same setup constraints as the first model. 150 replications and a one-hour warm-up period returned an average of 456 customers through the system. Similar to the first output, this model will be analyzed according to total time in system, queue length, queue wait times, and utilization.

The optimized model showed that the average customer was in the system for 1.988 minutes and the maximum time was 11.383 minutes. The queue waiting times and lengths are summarized below in Table 3.

Table 3: Queue Waiting Time and Length Data for Second Model

Queue	Waiting Time (Minutes)		Queue Length (Customers)	
	Average	Maximum	Average	Maximum
Waiting to Order 1	0.208	2.794	0.201	4
Waiting to Order 2	0.132	2.080	0.028	3
Waiting on Custom Item	0.271	7.372	0.029	4
Waiting to Checkout 1	0.387	4.594	0.402	6
Waiting to Checkout 2	0.105	2.311	0.025	4

The utilizations for all resources are listed in Table 4.

Table 4: Utilization Summary for Second Model

Resource	Utilization
Food Counter Clerk 1	0.7452
Food Counter Clerk 2	0.1712
Custom Order Clerk	0.1919
Cashier 1	0.5880
Cashier 2	0.1276

5. Summary of Findings

As predicted, implementing more resources and adjusting the setup helped to decrease average customer wait times and queue lengths. Table 5 shows the percentage improvements for parameters previously discussed.

Table 5: Percent Improvement Summary

Parameter	Improvement (%)
Total System Time Average	61.92
Total System Time Maximum	67.48
Waiting to Order Average Queue Length	92.88
Waiting to Order Average Wait Time	86.38
Waiting to Checkout Average Queue Length	93.36
Waiting to Checkout Average Wait Time	90.10

All queue lengths and average wait times were decreased significantly. The exception however, was the custom order queue. The change in queue length and wait time was not significant, because this process was not adjusted, and any changes can therefore be attributed to distributional differences.

Another notable difference was the change in utilization; while the goal of the second model was to more evenly distribute resource allocation, it was found that utilizations were not as even. In hindsight, however, this makes sense. We essentially changed our setup from a G/G/m system to a G/G/1 system with m lines, and although this makes lines shorter, it is found that capacity becomes uneven. Another observation is that the system improvement likely came from additional resources, rather than change in setup. From this observation, recommendations will be made to take an alternative approach which will not only decrease queue lengths and waiting times, but also balance utilization and possibly use less resources.

6. Conclusions and Lessons Learned

Using a version of the suggested system setup, improvements to the processing time, queuing time, and queue length would be expected to the revised system. Based on the analysis generated above and time spent from the team, we can conclude there are improvements to be made to the current process and system. Simulation has provided a venue to strategically and economically test different ways to improve Chick-fil-A's setup and operations. While there are many lessons learned, the biggest would include verifying the parameters set within the simulation software so that the time (route) a part or customer takes, does not actually take 1 hour if when, in reality, the process takes 1 second. This would create a huge difference in the reporting and data side of sale. Another action to consider would include the application of lean tools and kaizen to the current process/setup with Value Stream Mapping (VSM) for the most important value-added products. This will enable to let the stakeholders to see where they are currently positioned and create a "Future State" in which they can implement the desired value, process and metrics.

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